



2019 HSC Course Assessment Task 3
(Trial Examination)
Thursday June 27, 2019.

- Working time –3 hours (plus 5 minutes reading time).
- Write using blue or black pen. Diagrams may be sketched in pencil.
- Board approved calculators may be used.
- All necessary working should be shown in every question.
- Attempt all questions.

- Mark your answers on the answer sheet provided.

- Commence each new question on a new page.
- Show all necessary working in every question. Marks may be deducted for illegible or incomplete working.

BOOKLETS USED:

Ms Ziaziaris



Mr Lin

[illegible]

Section 1: Multiple Choice– 1 mark each.

Q1. What values for a and b satisfy the following equation:

$$\frac{14}{3 - \sqrt{2}} = a + \sqrt{b}$$

- (A) $a = 3$ and $b = 2$
- (B) $a = 6$ and $b = -2$
- (C) $a = 6$ and $b = 2$
- (D) $a = 6$ and $b = 8$

Q2. Which of the following is perpendicular to $2x - 4y + 1 = 0$?

- (A) $2x + y + 5 = 0$
- (B) $2x + 4y - 2 = 0$
- (C) $4x - 2y + 3 = 0$
- (D) $x - 2y - 1 = 0$

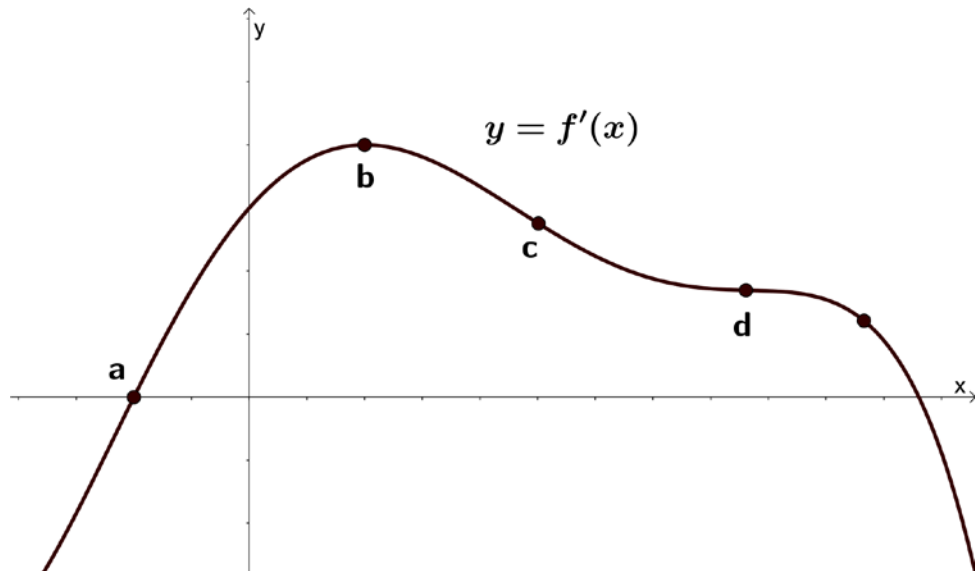
Q3. Which of the following is equal to $\frac{d}{dx} \log_e(\tan x)$?

- (A) $\cot x$
- (B) $\sec x \cos x$
- (C) $\frac{1}{\sec^2 x}$
- (D) $\sec^2 x \tan x$

Q4. Which expression is a factorisation of $343 - x^6$?

- (A) $(7 - x^2)(49 + 7x^2 + x^4)$
- (B) $(7 - x^2)(49 + 14x^2 + x^4)$
- (C) $(7 - x^2)(49 - 7x^2 + x^4)$
- (D) $(7 - x^2)(49 - 14x^2 + x^4)$

Q5. The diagram below shows the graph of the $y = f'(x)$, the derivative of a function:



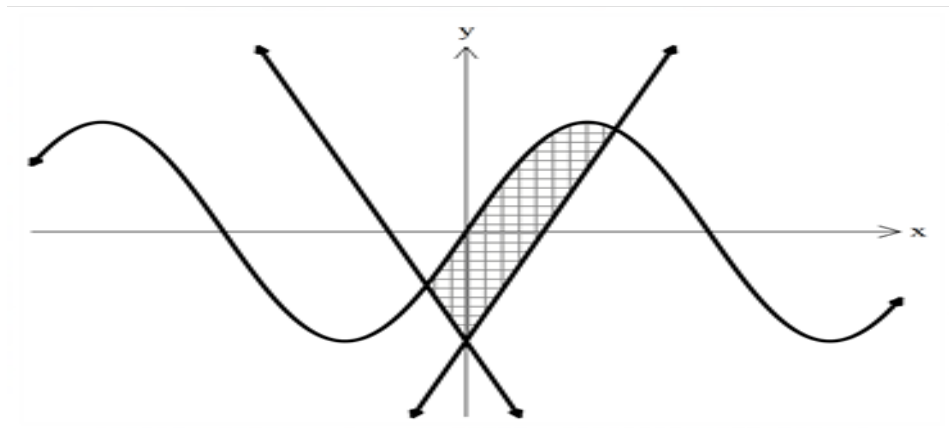
Which point on this graph indicates an inflexion point on $y = f(x)$?

- (A) a
- (B) b
- (C) c
- (D) d

Q6. The solution to $8x - 1 > 7x^2$ is

- (A) $\frac{1}{7} < x < 1$
- (B) $x < \frac{1}{7}, x > 1$
- (C) $x < -1, x > -\frac{1}{7}$
- (D) $-1 < x < -\frac{1}{7}$

Q7.



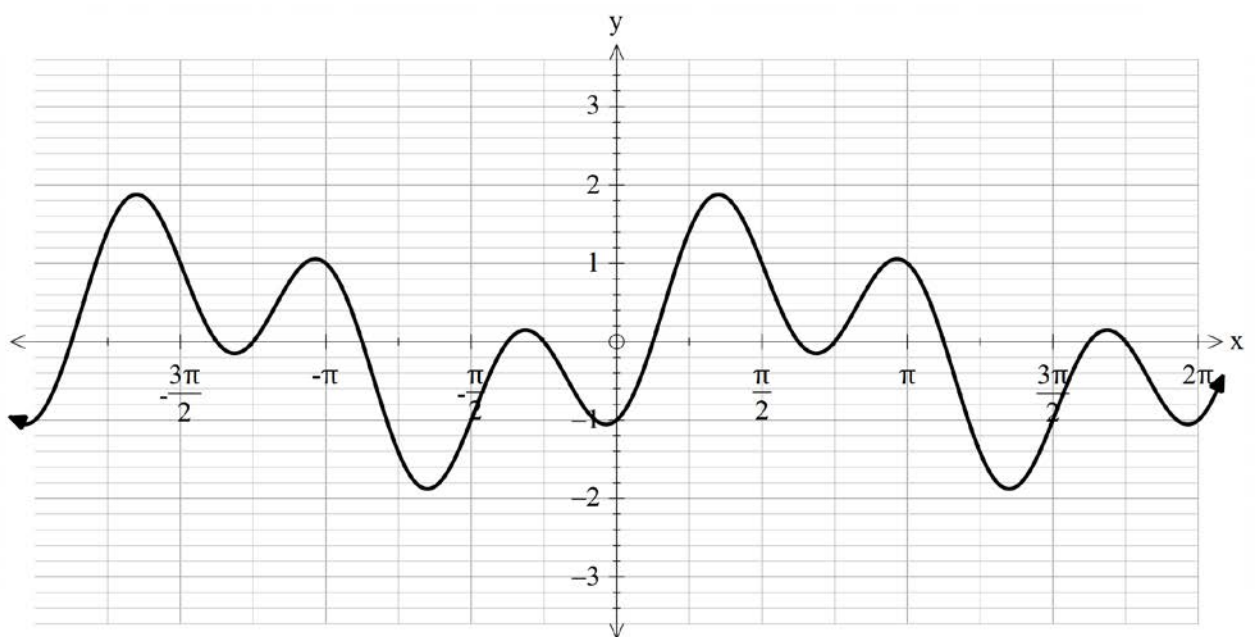
Which if the following best describes the shaded region?

- (A) $x - y - 1 \leq 0$ and $x + y + 1 \leq 0$ and $y \leq \sin x$
- (B) $x - y - 1 \leq 0$ and $x + y + 1 \geq 0$ and $y \leq \sin x$
- (C) $x - y - 1 \geq 0$ and $x + y + 1 \leq 0$ and $y \leq \sin x$
- (D) $x - y - 1 \geq 0$ and $x + y + 1 \geq 0$ and $y \leq \sin x$

Q8. Which of the following is equivalent to $\int_0^1 2^x dx$

- (A) $\frac{1}{\ln 2}$
- (B) $\frac{2}{\ln 2}$
- (C) $\ln 2$
- (D) $2 \ln 2$

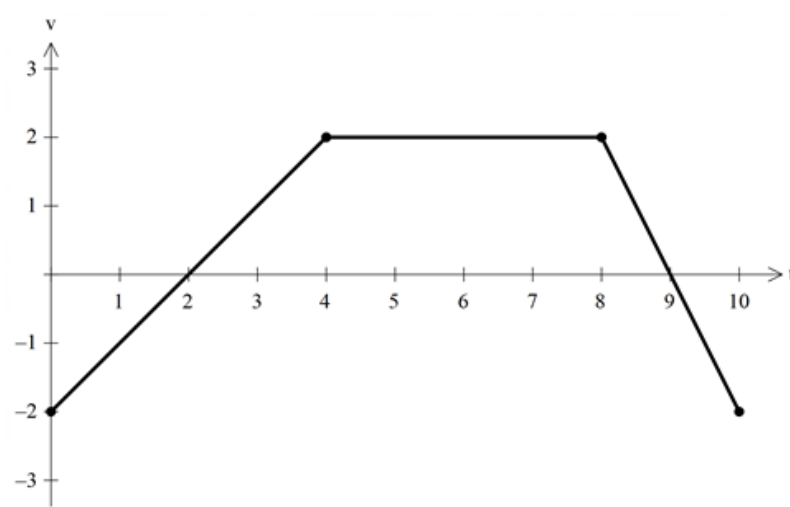
Q9. The graph of $y = \sin x - \cos 3x$ is shown:



How many solutions are there for the equation $2 \sin x - 2 \cos 3x = x$

- (A) 2
- (B) 3
- (C) 5
- (D) 7

- Q10. The following is a velocity-time diagram for a particle moving in one dimension, where time t is in seconds and velocity is in metres/second.



The particle is initially 2 metres to left of the origin.

When does the particle return to the origin?

- (A) 2 seconds
- (B) 4 seconds
- (C) 5 seconds
- (D) 6 seconds

End of Section I

Section II – Short Answer 90 marks

Question 11 (15 marks) Commence on a NEW page.

Marks

- (a) Factorise fully $2x^2y - 3x^2 - 8y + 12$ 2
- (b) State the domain and range of the function $f(x) = \sqrt{16 - x^2}$ 2
- (c) Solve $|3x - 1| = 4x$ 3
- (d) Differentiate $y = (e^{2x} + \log_e x)^4$ 2
- (e) Find the equation of tangent to the curve $y = 2 \cos x$ at $x = \frac{\pi}{3}$. 3
- (f) Solve $\cos 2x = \frac{\sqrt{3}}{2}$ for $0 \leq x \leq 2\pi$. 3

End of Question 11

Question 12 (15 marks) Commence on a NEW page.

Marks

(a) The points $A(-8, 4)$, $B(6, -2)$, $C(-4, 7)$ are defined in the Cartesian Plane.

- i) Show that the equation of the line AB is $6x + 14y - 8 = 0$. 1
- ii) Find the perpendicular distance from line AB to point C. 2
- iii) Hence or otherwise find the area of $\triangle ABC$. 2

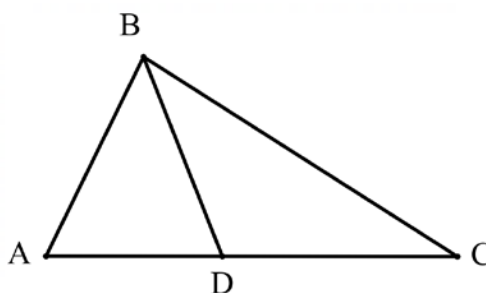
(b) Find:

- i) $\int \frac{dx}{(1 + 5x)^4}$ 1
- ii) $\int \sqrt{x} (4 - \sqrt{x}) dx$ 2
- iii) $\int_0^{\frac{\pi}{4}} \sin 2x dx$ 2

(c) Differentiate $y = \log_e \frac{5x + 1}{(x - 3)^2}$ 2

(d) Triangle ABC is an isosceles triangle with $AC = BC$.

The triangle was constructed so that $AB = BD = DC$.



- i) **Trace or copy the diagram into your workbook.**
- ii) Find the value of $\angle BCD$, showing reasoning. 3

End of Question 12

Question 13 (15 marks) Commence on a NEW page.

Marks

(a) For the parabola $12x + y^2 + 6y - 15 = 0$

i) Find the coordinates of the vertex.

2

ii) Find the coordinates of the focus.

2

(b) The roots of the equation $x^2 - 2kx + 3k = 0$ differ by four.

3

Find the value(s) of k .

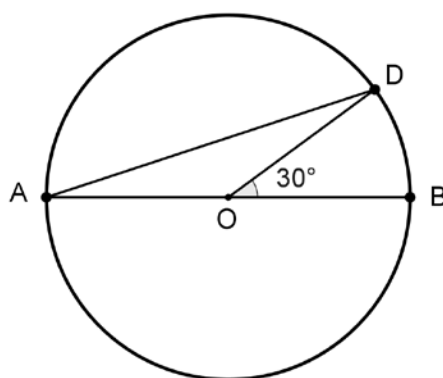
(c) A line L passes through the point $Q(4, -2)$ and the intersection of the lines

3

$$2x + 5y - 1 = 0 \text{ and } x + y - 5 = 0$$

Find the equation of L in general form.

(d) The diagram below shows a circle radius 1 unit, diameter AB and $\angle DOB = 30^\circ$



i) Explain why $\angle DAO = 15^\circ$.

1

ii) Show that $AD^2 = 2 + \sqrt{3}$.

2

iii) Hence show that $\sin 15^\circ = \frac{1}{2}\sqrt{2 - \sqrt{3}}$

2

End of Question 13

Question 14 (15 marks) Commence on a NEW page.

Marks

- (a) Given the function

$$f(x) = x^4 - 8x^3$$

- i) Find the stationary points of $y = f(x)$ and determine their nature.

4

- ii) Sketch the graph of $y = f(x)$.

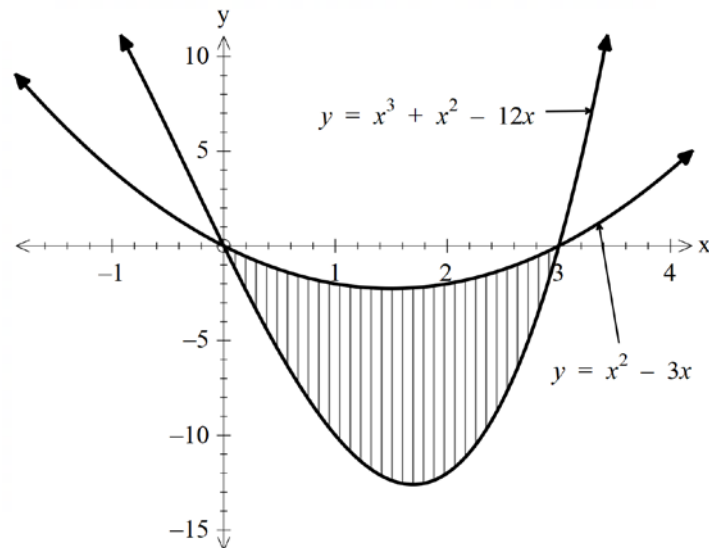
3

- (b) The diagram below shows the graphs of $y = x^2 - 3x$ and $y = x^3 + x^2 - 12x$.

3

The graphs intersect at (0,0) and (3,0).

Find the area between the two curves.



- (c) By considering the sum of a suitable infinite geometric series, express $0.54\dot{3}$ as a fraction in simplest form.

3

- (d) Evaluate $\sum_{k=3}^{100} (-1)^k (2k + 1)$.

2

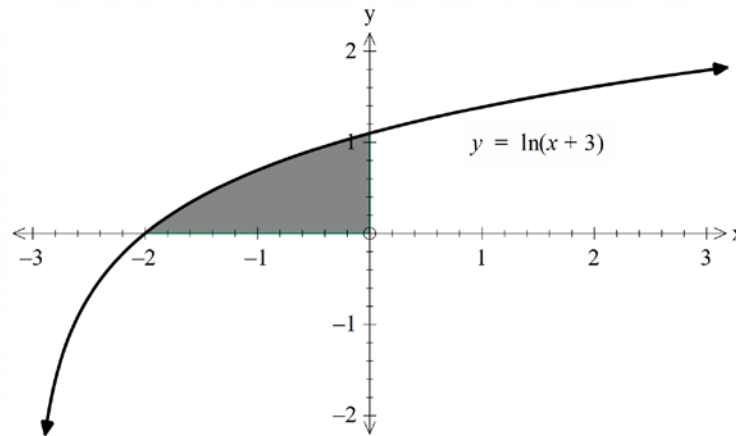
End of Question 14

Question 15 (15 marks) Commence on a NEW page.

- (a) The area enclosed by the curve $y = \log_e(x + 3)$, the x -axis and the y -axis is rotated about the y -axis.

4

Find the volume of the resulting solid of revolution, in exact form.



- (b) Consider the first quadrant of a circle radius 1:

4

$$y = \sqrt{1 - x^2}, \quad 0 \leq x \leq 1.$$

Use Simpson's Rule with 5 function values in the range $0 \leq x \leq 1$ to find an estimate for the area of the quadrant and hence show that

$$\pi \approx \frac{1}{3} (1 + \sqrt{3} + \sqrt{7} + \sqrt{15})$$

- (c) Use the fact that

2

$$\frac{u}{u-1} - \frac{u}{u+1} = \frac{2u}{u^2-1}$$

to find

$$\int \frac{e^x}{e^{2x} - 1} dx$$

Question 15 continues on the next page

Question 15 (continued)

(d) A particle travels along the x -axis with acceleration given by

$$\ddot{x} = -6t$$

where t is the time in seconds. The particle has an initial velocity of 9 m/s.

- | | | |
|------|---|---|
| i) | Find the equation of the velocity | 1 |
| ii) | When is the particle stationary? | 1 |
| iii) | What is the average speed in the first 4 seconds? | 3 |

End of Question 15

Question 16 (15 marks) Commence on a NEW page.

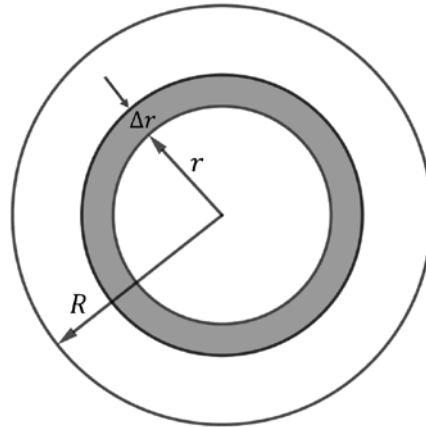
- (a) The YouTube channel “5 Minutes Physics” demonstrates how a circle can be dissected into thin strips, then transformed into a simple shape for which we can find the area.



We will now do a similar dissection using mathematics:

Consider a circle of radius R (see diagram below).

Within that circle, consider a thin annulus, r units from the centre, with a width Δr units.



- i) Draw a diagram to explain why the area of the annulus in the diagram above, when it is unrolled, can be expressed as: 1

$$A_{strip} \approx 2\pi r \cdot \Delta r$$

- ii) Using no more than three sentences, explain the ideas that allow us to develop the expression in part (i) into the following: 2

$$A_{circle} = \int_0^R 2\pi r \, dr$$

- iii) Hence derive the formula for the area of the circle. 1

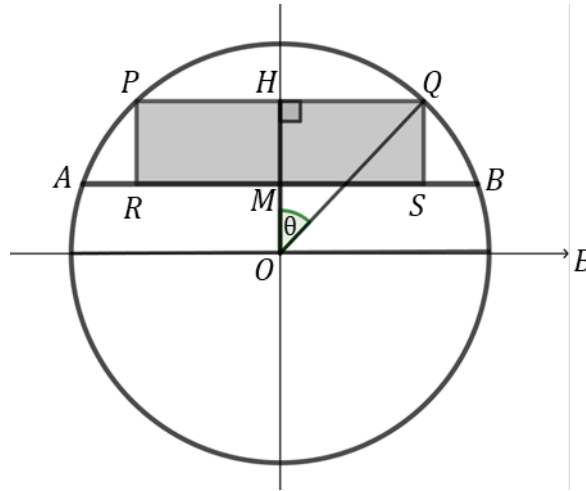
Question 16 continues on the next page

(b) In the diagram below, O is the centre of a circle radius R .

The chord AB has length kR and is parallel to the diameter, with midpoint M .

We now construct a rectangle inside the minor segment AB .

The angle θ is constructed as shown.



i) Show that $OM = R \sqrt{1 - \frac{k^2}{4}}$ 2

ii) Hence show that $p = \sqrt{1 - \frac{k^2}{4}}$ will always satisfy $0 < p < 1$. 1

iii) Show that the area of the rectangle is given by:

$$A = 2R^2 \sin \theta (\cos \theta - p) \quad \text{where} \quad p = \sqrt{1 - \frac{k^2}{4}} \quad \text{2}$$

iv) Show that $\frac{dA}{d\theta} = 2R^2(2 \cos^2 \theta - p \cos \theta - 1)$ 2

v) For this problem, show that $\frac{d^2A}{d\theta^2} < 0$ 2

vi) Hence show that the area of the rectangle will be maximized when 2

$$\cos \theta = \frac{1}{4} \left(p + \sqrt{p^2 + 8} \right)$$

END OF THE EXAMINATION.

Multiple Choice

5

1. (D) $\frac{14}{3-\sqrt{2}} = 6+2\sqrt{2} = 6+\sqrt{8}$

2. (A)

3. ☒ $\frac{\sec^2 x}{\tan x} = \sec x \csc x$

4. (A)

5. (B) Inflection when concavity changes sign.
Gradient of the curve in the diagram ($f'(x)$) is concavity $f''(x)$.
At Point B, gradient (ie: concavity) changes from +ve to -ve

6. (A)

7. (B)

8. (A)

9. (C) Construct the line $y = \frac{\pi}{2}$ and count intersections

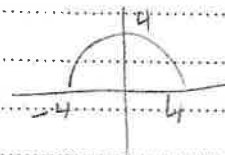
10. (E) Observe we start at $x = -2m$
From $t=0$ to $t=2$ we keep moving left,
 $t=2$ to $t=4$ return around at $t=0$
back at $x = -2m$. So at $x=0$ when $t=5$

17

Q4. a) $2x^2y - 3x^2 - 8y + 12$
 $= x^2(2y-3) - 4(2y-3)$
 $= (x^2-4)(2y-3)$
 $= (x+2)(x-2)(2y-3)$

(2)

b) $f(x) = \sqrt{16-x^2}$



D: $-4 \leq x \leq 4$

R: $0 \leq y \leq 4$

(12)

c) $|3x-1| = 4x$

(13)

$\therefore 3x-1 = 4x$ or $3x-1 = -4x$
 $x = -1$ $7x = 1$

But not valid.
 $-4 \neq -4x$

$x = \frac{1}{7}$
valid ✓

$\therefore x = \frac{1}{7}$

$\cos 2x = \frac{\sqrt{3}}{2}$ $0 \leq x \leq 2\pi$
 $0 \leq 2x \leq 4\pi$

$2x = \frac{\pi}{6}, \frac{11\pi}{6}, \frac{13\pi}{6}, \frac{23\pi}{6}$

$\therefore x = \frac{\pi}{12}, \frac{11\pi}{12}, \frac{13\pi}{12}, \frac{23\pi}{12}$

(13)

(2)

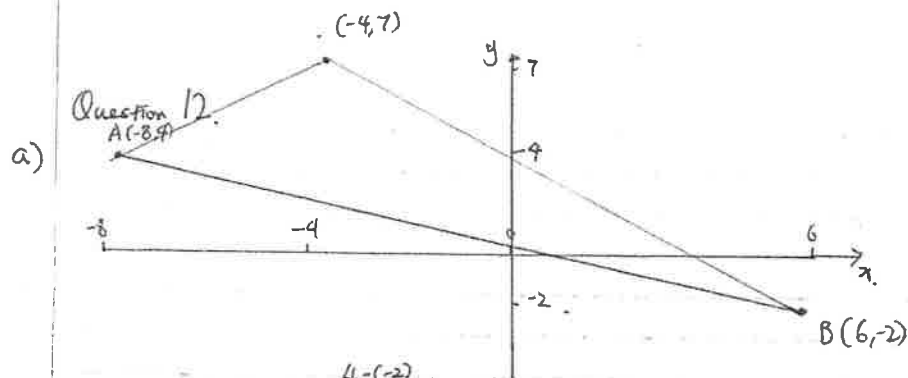
d) $y = (e^{2x} + \log x)^4$
 $y' = 4(e^{2x} + \log x)^3 (2e^{2x} + \frac{1}{x})$

(13)

e) $y = 2\cos x$ $(\frac{\pi}{3}, 1)$
 $y' = -2\sin x$
 $y'(\frac{\pi}{3}) = -2\sin \frac{\pi}{3}$
 $= -2 \cdot \frac{\sqrt{3}}{2} = -\sqrt{3}$

$\therefore$ tangent: $(y-1) = -\sqrt{3}(x-\frac{\pi}{3})$ ✓✓

or $y = -\sqrt{3}x - \frac{\sqrt{3}\pi}{3} + 1$



i) $AB: y+2 = \frac{4-(-2)}{-8-6}(x-6)$

$$y+2 = \frac{6}{-14}(x-6)$$

$$-14y-28 = 6x-36$$

$$0 = 6x+14y-8 \quad \checkmark \quad \text{as required.}$$

ii) $\perp d = \frac{|ax+by+c|}{\sqrt{a^2+b^2}} = \frac{|6(-4)+14(7)-8|}{\sqrt{6^2+14^2}} \quad \checkmark$

$$= \frac{66}{\sqrt{232}} \quad \checkmark = \frac{33}{\sqrt{58}} \quad (2)$$

iii) Area $\Delta = \frac{1}{2}bh$

$$= \frac{1}{2} \times \sqrt{(6-(-8))^2 + (-2-4)^2} \times \frac{66}{\sqrt{232}} \quad \checkmark$$

$$= \frac{1}{2} \times \sqrt{232} \times \frac{66}{\sqrt{232}} \quad \checkmark = 33 \quad (2)$$

b) i) $\int \frac{1}{(1+5x)^4} dx = \frac{1}{5} \int \frac{5}{(1+5x)^4} dx$

$$= \frac{1}{5} \times \frac{-1}{3(1+5x)^3} + C$$

$$= \frac{-1}{15(1+5x)^3} + C \quad \checkmark \quad (1)$$

ii) $\int \sqrt{x}(4-x) dx = \int 4x^{\frac{1}{2}} - x dx \quad \checkmark$

$$= 4 \times \frac{2}{3} x^{\frac{3}{2}} - \frac{1}{2} x^2 + C$$

$$= \frac{8}{3} x^{\frac{3}{2}} - \frac{1}{2} x^2 + C \quad \checkmark \quad (2)$$

iii) $\int_0^{\frac{\pi}{2}} \sin 2x dx = \left[-\frac{\cos 2x}{2} \right]_0^{\frac{\pi}{2}} \quad \checkmark$

$$= -\frac{\cos \pi}{2} - \frac{-\cos 0}{2} = 0 + \frac{1}{2} = \frac{1}{2} \quad \checkmark \quad (2)$$

c) $y = \ln \left(\frac{5x+1}{(x-3)^2} \right)$ let $u = \frac{5x+1}{(x-3)^2}$

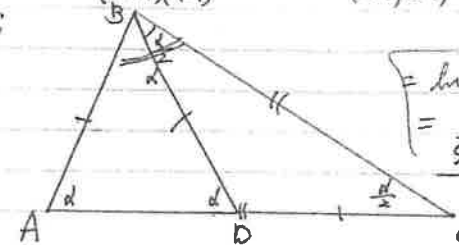
$$y = \ln u$$

$$\frac{dy}{du} = \frac{1}{u} = \frac{(x-1)^2}{5x+1}$$

$$\frac{dy}{dx} = \frac{(x-1)^2}{5x+1} \times \frac{-5x+1}{(x-1)^3}$$

$$= \frac{-5x+1}{(5x+1)(x-1)} = -\frac{5x+1}{(5x+1)(x-1)} = -\frac{5x+1}{5x^2+4x-1} \quad \checkmark \quad (2)$$

d) i



$$= \ln(5x+1) - 2 \ln(x-3)$$

$$= \frac{5}{5x+1} - \frac{2}{x-3}$$

ii) let $\angle BAD = d$

$$\angle BDC = 180-d \quad (\text{straight line}) \quad \checkmark$$

$$\angle DBC = \angle BCD = \frac{180-(180-d)}{2} = \frac{d}{2} \quad (\text{base angles of isosceles triangle are equal})$$

$$\angle CBA = \angle BAC = d \quad (\text{base angles of isosceles triangle are equal})$$

$$d+d+\frac{d}{2} = 180 \quad (\text{angle sum of a triangle equals to } 180^\circ)$$

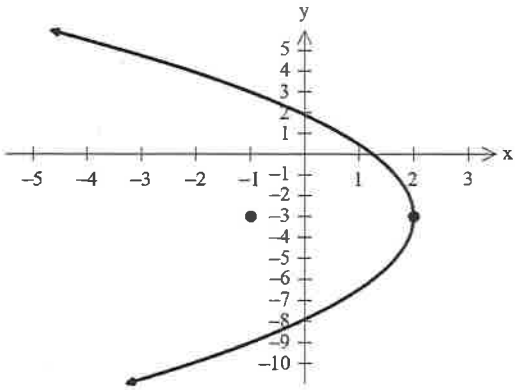
$$\therefore \frac{5d}{2} = 180^\circ$$

$$5d = 360^\circ$$

$$d = \frac{360^\circ}{5} = 72^\circ \quad \checkmark$$

$$\therefore \angle BCD = \frac{d}{2} = \frac{72}{2} = 36^\circ$$

(3)

Solutions to Question 13 – 2 Unit Advanced Task 3 2019		
(a)	$12x + y^2 + 6y - 15 = 0$ $y^2 + 6y = -12x + 15$ $y^2 + 6y + 9 = -12x + 15 + 9$ $(y + 3)^2 = -12x + 24$ $(y + 3)^2 = -12(x - 2)$ $(y + 3)^2 = -4(3)(x - 2)$  (i) Vertex = (2, -3) (ii) Focus = (-1, -3)	✓ Putting the equation into the appropriate form ✓ Finding the vertex ✓ Finding the focal length ✓ Finding the focus
(b)	Let the roots be $(p - 2)$ and $(p + 2)$ $(p + 2) + (p - 2) = -\frac{b}{a}$ $2p = -\frac{-2k}{1}$ $2p = 2k$ $p = k \dots \textcircled{1}$ $(p + 2)(p - 2) = \frac{c}{a}$ $p^2 - 4 = \frac{3k}{1}$ $p^2 - 4 = 3k$ $k^2 - 4 = 3k \text{ from } \textcircled{1}$ $k^2 - 3k - 4 = 0$ $(k + 1)(k - 4) = 0$ $\therefore k = -1 \text{ or } k = 4$	✓ Creating an equation with the roots ✓ Finding an equation in terms of k only ✓ Solving to find both values of k

(c)	$(2x + 5y - 1) + k(x + y - 5) = 0$ $(2(4) + 5(-2) - 1) + k(4 + (-2) - 5) = 0$ $(8 - 10 - 1) + k(4 - 2 - 5) = 0$ $-3 - 3k = 0$ $3k = -3$ $k = -1$ $(2x + 5y - 1) + (-1)(x + y - 5) = 0$ $2x + 5y - 1 - x - y + 5 = 0$ $x + 4y + 4 = 0$	✓ Writing an equation with the k method ✓ Solving to find k ✓ Finding the equation of the line in general form
	Alternative solution $2x + 5y - 1 = 0 \dots \textcircled{1}$ $x + y - 5 = 0 \dots \textcircled{2}$ $(1) - (2) \times 2$ $2x + 5y - 1 = 0 \quad -$ $2x + 2y - 10 = 0$ $3y + 9 = 0$ $3y = -9$ $y = -3$ sub into $\textcircled{2}$ $x + (-3) - 5 = 0$ $x - 3 - 5 = 0$ $x - 8 = 0$ $x = 8$ point of intersection is (8, -3) $\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$ $\frac{y - (-2)}{x - 4} = \frac{-3 - (-2)}{8 - 4}$ $\frac{y + 2}{x - 4} = -\frac{1}{4}$ $4(y + 2) = -1(x - 4)$ $4y + 8 = -x + 4$ $x + 4y + 4 = 0$	✓ Finding the point of intersection ✓ Using the two point/point gradient formula ✓ Finding the equation of the line in general form

(d)	(i)	$\angle DAO, \angle ADO$ and $\angle DOB$ all refer to the acute angles $\angle DAO + \angle ADO = \angle DOB$ (exterior angle sum of $\triangle AOD$) but as $\triangle AOD$ is isosceles $\angle DAO = \angle ADO$ $\therefore 2\angle DAO = \angle DOB$ $2\angle DAO = 30^\circ$ $\angle DAO = 15^\circ$	✓ Finding $\angle DAO$ with geometrical reasons stated
	(ii)	$\angle AOD$ and $\angle DOB$ refer to the obtuse and acute angles respectively $\angle AOD = 180^\circ - \angle DOB$ (angles on a straight line are supplementary) $\angle AOD = 180^\circ - 30^\circ$ $= 150^\circ$ $AD^2 = AO^2 + DO^2 - 2 \times AO \times DO \times \cos(\angle AOD)$ $= 1^2 + 1^2 - 2 \times 1 \times 1 \times \cos(150^\circ)$ $= 2 - 2 \times \left(\frac{-\sqrt{3}}{2} \right)$ $= 2 - (-\sqrt{3})$ $= 2 + \sqrt{3}$	✓ Using the cosine rule with appropriate sides and angles ✓ Clear substitutions and steps to arrive at the answer
	(iii)	$\angle DAO$ and $\angle AOD$ refer to the acute and obtuse angles respectively $\frac{\sin(\angle DAO)}{DO} = \frac{\sin(\angle AOD)}{AD}$ $\frac{\sin(15^\circ)}{1} = \frac{\sin(150^\circ)}{\sqrt{2+\sqrt{3}}}$ $\left(\frac{1}{2} \right)$ $\sin(15^\circ) = \frac{1}{\sqrt{2+\sqrt{3}}}$ $= \frac{1}{2} \times \frac{1}{\sqrt{2+\sqrt{3}}} \times \frac{\sqrt{2-\sqrt{3}}}{\sqrt{2-\sqrt{3}}}$ $= \frac{1}{2} \times \frac{\sqrt{2-\sqrt{3}}}{\sqrt{2^2 - (\sqrt{3})^2}}$ $= \frac{1}{2} \times \frac{\sqrt{2-\sqrt{3}}}{\sqrt{4-3}}$ $= \frac{1}{2} \times \frac{\sqrt{2-\sqrt{3}}}{1}$ $= \frac{1}{2} \sqrt{2-\sqrt{3}}$	✓ Using the sine rule with correct values ✓ Clearly show the steps (including rationalising the denominator) leading to the answer

Q14(a) $f(x) = x^4 - 8x^3$
 $f'(x) = 4x^3 - 24x^2$
 $= 4x^2(x-6)$

$f''(x) = 12x^2 - 48x$

stationary points at $f'(x) = 0$

ie $4x^2(x-6) = 0$

$x = 0, x = 6$

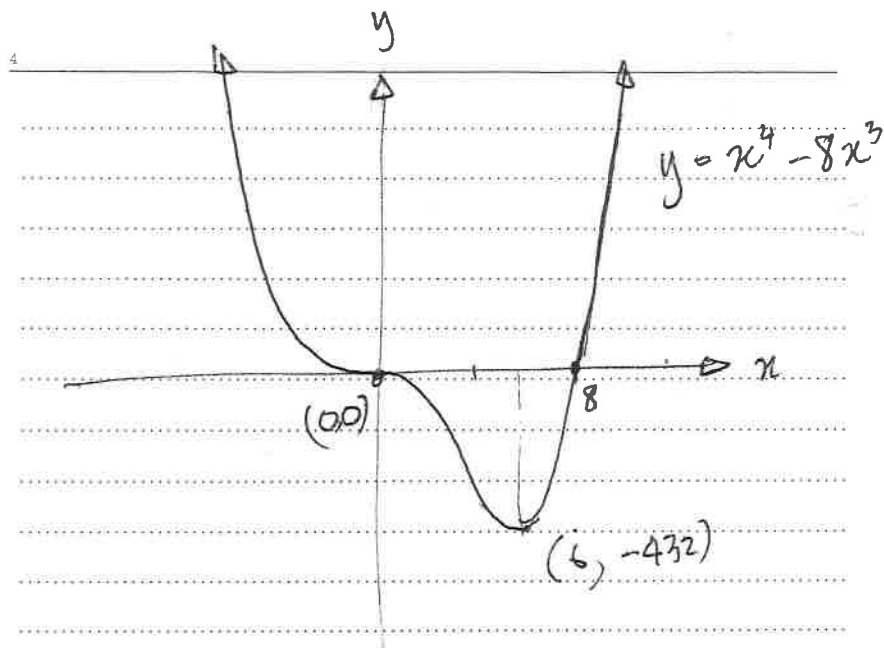
$(0,0), (6,-432)$

Testing $(0,0)$: $f''(0) = 0 \leftarrow$ Not a valid test!

x	-1	0	3	6	7
$f'(x)$	28	0	-108	0	196
	\	-	\	-	/

$\therefore (0,0)$ is a horizontal point of inflexion

$\therefore (6,-432)$ is local minimum.



[Optimal: identify (4, -256) as another inflexion point - but not required]

$$\begin{aligned}
 14 \text{ b) } A &= \int_0^3 ((x^2 - 3x) - (x^3 + x^2 - 12x)) dx \\
 &= \int_0^3 (-3x - x^3 + 12x) dx \\
 &= \int_0^3 (-x^3 + 9x) dx \\
 &= \left[\frac{9x^2}{2} - \frac{x^4}{4} \right]_0^3 = \left(\frac{9(9)}{2} - \frac{81}{4} \right) - (0 - 0) \\
 &= \frac{81}{4} u^2
 \end{aligned}$$

$$14 \text{ c) } 0.5\overline{43} = 0.5 + 0.043 + 0.00043 + \dots + \dots$$

$$= \frac{1}{2} + \frac{43}{1000} + \frac{43}{1000} \left(\frac{1}{100} \right) + \frac{43}{1000} \left(\frac{1}{100} \right)^2 + \dots$$

$$\text{GP } a = \frac{43}{1000}, r = \frac{1}{100}$$

$$= \frac{1}{2} + \frac{\frac{43}{1000}}{1 - \frac{1}{100}} \quad (S_{\infty} = \frac{a}{1-r} \quad |r| < 1)$$

$$= \frac{1}{2} + \frac{43}{1000} \times \frac{100}{99}$$

$$= \frac{269}{495}$$

$$\begin{aligned}
 \text{d) } \sum_{k=3}^{100} (-1)^k (2k+1) &= (-1)^{2(3)+1} + (-1)^{2(4)+1} \\
 &+ (-1)^{2(5)+1} + (-1)^{2(6)+1} \\
 &+ \dots + (-1)^{2(99)+1} + (-1)^{2(100)+1}
 \end{aligned}$$

$$\begin{aligned}
 n &= 100 - 3 + 1 \\
 &= 96
 \end{aligned}$$

$$\begin{aligned}
 &= -7 + 9 \\
 &- 11 + 13 \\
 &+ \dots - 199 + 200
 \end{aligned}$$

$$= 2 + 2 + 2 + \dots + 2$$

96/2 pairs

$$= \frac{96}{2} \times 2 = 96$$

Question 15

a)

$$y = \ln(x+3)$$

$$e^y = x+3, \quad x = e^y + 3$$

$$V = \pi \int_0^{\ln 3} (e^y + 3)^2 dy$$

$$= \pi \int_0^{\ln 3} e^{2y} + 6e^y + 9 dy$$

$$= \pi \left[\frac{e^{2y}}{2} + 6e^y + 9y \right]_0^{\ln 3}$$

$$= \pi \left[\left(\frac{1}{2} e^{4 \ln 3} + 6e^{\ln 3} + 9 \ln 3 \right) - \left(\frac{1}{2} + 6 + 0 \right) \right]$$

$$= \pi \left(\frac{9}{2} - 18 + 9 \ln 3 - \left(\frac{1}{2} + 6 \right) \right)$$

$$= \pi (9 \ln 3 - 8) u^3$$

b).

$$y = \sqrt{1-x^2} \quad 0 \leq x \leq 1$$

x	0	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$	1
y	1	$\frac{\sqrt{15}}{4}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{7}}{4}$	0

$$h = \frac{1}{4}$$

$$A \approx \frac{1}{12} \left[1 + 0 + 4 \left(\frac{\sqrt{15}}{4} + \frac{\sqrt{7}}{4} \right) + 2 \left(\frac{\sqrt{3}}{2} \right) \right]$$

$$= \frac{1}{12} (1 + \sqrt{15} + \sqrt{7} + \sqrt{3}) u^2$$

$$A = \frac{1}{4} \pi r^2 = \frac{1}{4} \pi u^2$$

$$\frac{\pi}{4} \approx \frac{1}{12} (1 + \frac{\sqrt{3}}{3} + \sqrt{7} + \sqrt{15})$$

$$\therefore \pi \approx \frac{1}{3} (1 + \sqrt{3} + \sqrt{7} + \sqrt{15})$$

c)

$$\frac{e^x}{e^x - 1} - \frac{e^x}{e^x + 1} = \frac{2e^x}{e^{2x} - 1} = 2 \times \frac{e^x}{e^{2x} - 1} \quad (\text{given})$$

$$\therefore \int \frac{e^x}{e^{2x} - 1} dx = \frac{1}{2} \int \frac{2e^x}{e^{2x} - 1} dx$$

$$= \frac{1}{2} \int \frac{e^x}{e^x - 1} - \frac{e^x}{e^x + 1} dx$$

$$= \frac{1}{2} [\ln(e^x - 1) - \ln(e^x + 1)] + c$$

$$= \frac{\ln(e^x - 1)}{2} - \frac{\ln(e^x + 1)}{2} + c$$

d)

Question 15

$$\ddot{x} = -6t$$

$$\dot{x}(0) = 9 \text{ m/s}$$

$$\text{i) } \dot{x} = \int \ddot{x} dt$$

$$= -3t^2 + C$$

$$9 = -3(0) + C \quad \therefore C = 9$$

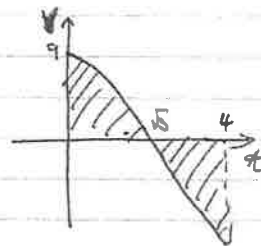
$$\therefore \dot{x} = -3t^2 + 9$$

$$\text{ii) when } \dot{x} = 0$$

$$9 - 3t^2 = 0$$

$$t = \sqrt{3} \text{ s } (t > 0)$$

iii)



$$\int_0^{\sqrt{3}} 9 - 3t^2 dt + \int_{\sqrt{3}}^4 9 - 3t^2 dt$$

= distance

$$= \left| [9t - t^3]_0^{\sqrt{3}} \right| + \left| [9t - t^3]_{\sqrt{3}}^4 \right|$$

$$= (9\sqrt{3} - 3\sqrt{3}) + [(36 - 64) - 6\sqrt{3}]$$

$$= 6\sqrt{3} + [-28 - 6\sqrt{3}]$$

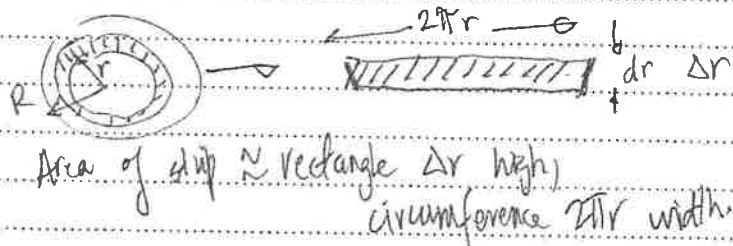
$$= 12\sqrt{3} + 28$$

$$\text{average speed} = \frac{12\sqrt{3} + 28}{4} = 3\sqrt{3} + 7 \text{ m/s}$$

Q16.9

(1)

(i)

(ii) Area whole circle: add the strips from $r=0$ to $r=R$

$$A \approx \sum_{r=0}^{r=R} (2\pi r) \Delta r$$

(2) to get exact shrink to very small (infinitesimal / limit)
 $\Delta r \rightarrow dr$

$$\therefore A = \int_0^R 2\pi r \, dr$$

$$(iii) \therefore A = \int_0^R 2\pi r \, dr$$

(3)

$$\propto 2\pi$$

$$= [\pi r^2]_0^R$$

$$= \pi R^2 - 0$$

$$= \pi R^2 \quad (u2)$$

Q16(b) *DRAW DIAGRAM*

(i)

By definition, $MB = \frac{k}{2} R$ By Pythagoras $R^2 = \left(\frac{k}{2} R\right)^2 + OM^2$

$$\therefore OM = \sqrt{R^2 - \left(\frac{kR}{2}\right)^2} \quad (OM > 0)$$

$$= R \sqrt{1 - \frac{k^2}{4}}$$

(ii) By definition $OM = pR$ from diagram $0 < OM < R$

$$\therefore 0 < pR < R$$

$$\therefore 0 < p < 1$$

or you could use algebra (more work, less insight!)

$$0 \leq k \leq 2 \quad (\text{from diagram})$$

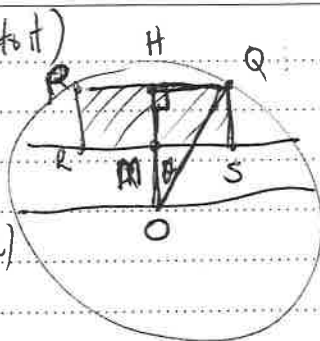
$$0 \leq k^2 \leq 4$$

$$0 \leq \frac{k^2}{4} \leq 1$$

$$\therefore 0 \leq 1 - \frac{k^2}{4} \leq 1$$

$$\therefore 0 \leq \sqrt{1 - \frac{k^2}{4}} \leq 1$$

(iii) DRAW DIAGRAM (or refer to it)



$$HO = R \cos \theta$$

$$HM = HO - OM$$

$$\therefore HM = R \cos \theta - pR \quad (\text{from ii})$$

$$\therefore \text{width } PQ = 2(R \sin \theta)$$

$$\therefore \text{Area} = (2R \sin \theta)(R \cos \theta - pR)$$

$$= 2R^2 \sin \theta (\cos \theta - p) \quad \text{as required.}$$

$$(iv) \frac{dA}{d\theta} = 2R^2 (\cos \theta (\cos \theta - p) + \sin \theta (-\sin \theta))$$

$$= 2R^2 (\cos^2 \theta - p \cos \theta - \sin^2 \theta)$$

$$= 2R^2 (\cos^2 \theta - p \cos \theta - (1 - \cos^2 \theta))$$

$$= 2R^2 (2 \cos^2 \theta - p \cos \theta - 1)$$

$$(v) \frac{d^2 A}{d\theta^2} = 2R^2 (-4 \sin \theta \cos \theta + p \sin \theta)$$

$$= 2R^2 \sin \theta (p - 4 \cos \theta)$$

$$\text{Consider } 2R^2 \sin \theta (p - 4 \cos \theta)$$

$$\sin \theta > 0 \text{ for } 0 < \theta < \frac{\pi}{2}$$

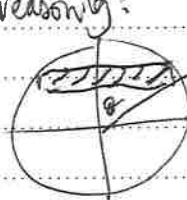
$$\text{From the area of the rectangle: } \cos \theta - p > 0$$

$$\therefore p - \cos \theta < 0$$

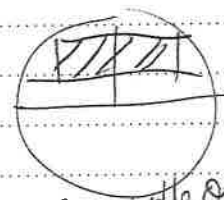
$$\therefore p - 4 \cos \theta < 0$$

$$\text{so } \frac{d^2 A}{d\theta^2} = 2R^2 \text{ is } (+)(-) < 0 \text{ for } 0 < \theta < \frac{\pi}{2}$$

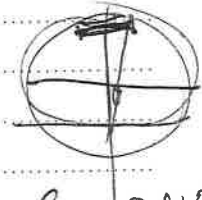
Alternate reasoning:



for large θ , $A < 0$

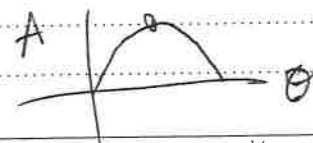


for middle θ ,
 $A > 0$



for $\theta \rightarrow 0$,
 $A < 0$

so area must do this:



ie there is a maximum.

[should also say $A(\theta)$ is continuous]

(v) Solve the equation $\frac{dA}{d\theta} = 0$

$$2R^2(2\cos^2\theta - p\cos\theta - 1) = 0$$

ie $\cos\theta = \frac{p \pm \sqrt{p^2 + 8}}{4}$ (quadratic in disguise)

$\cos\theta$ must be positive ($0 \leq \theta \leq \frac{\pi}{2}$)

But $p - \sqrt{p^2 + 8}$ is negative; (must show why)

(a) $0 \leq p \leq 1$ or $p^2 < p^2 + 8$
 $\therefore 1 - \sqrt{1^2 + 8} < 0$
 $0 - \sqrt{0^2 + 8} < 0$
 $\sqrt{p^2} < \sqrt{p^2 + 8}$
 $p < \sqrt{p^2 + 8}$ for $p > 0$

only positive result is:

$$\cos\theta = \frac{1}{4}(p + \sqrt{p^2 + 8})$$